

10 TIPS FOR COOLING

Now that your system is up and running, it is important that you keep it cool for greater reliability. A cool system is also better equipped to handle overclocking experiments

1 Case cooling fans

If your casing has provisions for cooling fans, invest in a couple of good ones. A CPU fan concentrates on only the processor and additional system cooling fans draw hot air from heat-generating components such as the CD-ROM, chipset and graphics card, maintaining a constant airflow within the system cabinet.

2 Upgrade your processor cooling

While the bundled heatsink and CPU fan do a good job of cooling the CPU during normal operations, you will need a more robust solution if your system is situated in an especially warm environment or while over-

clocking. In such cases, choose a more powerful solution from brands such as ThermalTake or CoolerMaster. Also, see to it that the CPU fan has an airflow of at least 34 cfm (cubic feet per minute) for reliable cooling under strenuous thermal conditions.

3 Of frequencies and voltages

While installing your motherboard and processor, make sure that your voltage and frequency settings are at their correct values, unless you mean to be overclocking them. An incorrect setting could run sections of your system with higher voltage

and frequency ratings than they are meant to, resulting in higher heat levels. Therefore, consult the motherboard manual while setting the voltages and frequencies.

4 Get a system dashboard

Most systems have applications that monitor the sensor chip on your motherboard to keep you informed about the current temperatures, frequencies and voltages of its critical components. This is especially useful while overclocking, when you want to keep the rated thermal levels under safe limits.

5 Arrange your cables

Internal cables such as your IDE, power and sensor cables can cause a mess within the system and hamper the airflow. Make sure that these cables are correctly ordered and tied together to circulate the air within the cabinet for more efficient cooling.

6 Double-barrel cooling

Opt for a casing that has at least two blowholes with provisions for cooling fans. One fan should be placed in the front and the other in the rear of the system for smooth circulation.

7 Additional cooling

If you are going to carry out overclocking experiments, make space for additional cooling for your processor, graphics card,

RAM and motherboard by modifying your system casing and installing more fans.

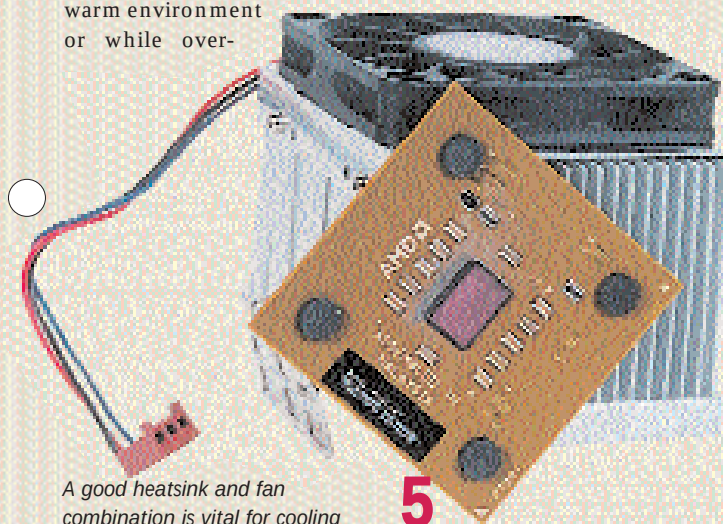
8 Go with the brushless
Invest in a brushless type of cooling fan as these are far more efficient—they are silent and feature a higher rpm as compared to older fans.

9 Clean the fans regularly

Case cooling fans are susceptible to picking up dust and dirt from the environment and need to be taken care of. If they are too clogged with dust, they could slow down or even stop rotating completely, causing overheating or even failure of components. Clean out these fans with a brush or a blower so as to dislodge any build-up of dust on the fan blades. It is wise to keep your system covered during the day when it is not being used as this prevents dust from settling on it.

10 Keep the system away from dusty environments to prevent dirt clogging

Keep your system away from dusty environments—this includes open windows facing main roads or construction sites. While a sealed, air conditioned room is best suited to a computer, today's computers are hardy enough to work well in a normal room, but care needs to be taken that they are not in direct sunlight, causing unnecessary build-up of heat.



A good heatsink and fan combination is vital for cooling your processor